

# Factors influencing university students' intention to use and reliance on generative artificial intelligence: An extended technology acceptance model with critical use

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## ARTICLE INFO

### Keywords:

Technology acceptance model  
Partial least squares structural equation modelling  
Generative artificial intelligence  
Critical use  
Reliance

## ABSTRACT

In light of the growing popularity, accessibility, and utility of Generative Artificial Intelligence (GenAI) tools, their impact on learning can be either supportive or detrimental, depending on how students use them. This calls for an examination of students' intention to use and reliance on GenAI to inform timely interventions within higher education to ensure favourable learning outcomes. This study investigates factors influencing intentions to use and reliance on GenAI among Engineering and Computer Science/Information Technology (CS/IT) students. Drawing on an extended Technology Acceptance Model (TAM), the study integrates the construct of critical use and conceptualises reliance across four functional domains relevant to engineering and CS/IT education. Data from an anonymous survey collected at an Australian university ( $n = 126$ ) are analysed using the Partial Least Squares Structural Equation Modelling (PLS-SEM) approach. The modelling results show that attitudes towards GenAI and critical use directly and positively influence intention to use GenAI, while perceived ease of use and perceived usefulness have positive indirect effects. The intention to use, in turn, significantly predicts reliance on GenAI across the four domains: understanding, assessment, programming, and engineering projects. Students demonstrate a moderate level of reliance overall, with greater use for understanding-related tasks and limited utilisation for full assessment writing. This study offers insights for higher education institutions aiming to foster ethical and critical use of GenAI for learning.

## 1. Introduction

Over the past few years, the rapid advancement of Artificial Intelligence (AI) has profoundly transformed nearly every sector worldwide, with Generative Artificial Intelligence (GenAI) at the forefront of this technological transformation. Its ubiquitous applications span business automation, personalised healthcare, smart environments, and education. In higher education, its rapid integration has reshaped pedagogy, classroom dynamics, and assessments through enhanced personalisation and interactivity (Hutson et al., 2022; Walczak & Cellary, 2023).

The educational benefits of GenAI are widely recognised, including instant and personalised feedback, multilingual accessibility, clarification of complex concepts, and improved learner engagement (Hazzan & Erez, 2025; Lee et al., 2025). In engineering education, in particular, GenAI could support design thinking, embedded systems, and

programming courses, which offer guidance, debugging assistance and iterative learning opportunities (Pham et al., 2023). Banerjee et al. (2025) also found that GenAI, specifically ChatGPT, facilitates programming tasks by generating code and explanations in computing and engineering education. Its benefits are further highlighted in these fields as a tool for information access, summarisation, revision, and cybersecurity assessment.

Nevertheless, in more complex domains such as computer architecture, users need solid subject expertise to fully harness GenAI's potential. Moreover, the availability of GenAI can reduce students' motivation for independent problem-solving (Hazzan & Erez, 2025). Growing concerns have also emerged regarding ethical use, data privacy, content bias, feedback reliability, and over-reliance on automated systems (Ariza et al., 2025; Zou & Huang, 2024). While some universities have opted to restrict or ban their use to protect academic integrity (Johnson,

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<https://doi.org/10.1016/j.caeai.2026.100618>

Received 15 December 2025; Received in revised form 20 April 2026; Accepted 23 May 2026

Available online 25 May 2026

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2023), others view them as catalysts for reimagining traditional pedagogy and assessment methods (Bernabei et al., 2023). As Campbell (2023) mentioned, GenAI is here to stay; thus, the challenge for educators and learners is not to resist or replace it but to develop critical, collaborative, and reflective approaches that ensure technology enhances rather than undermines human cognition and creativity. While GenAI continues to transform teaching and learning (Chen et al., 2024), students remain at the forefront of this change as early adopters. Understanding and predicting their intention to use GenAI is therefore vital. It provides valuable insights into emerging behavioural patterns, readiness for responsible use, and potential barriers to meaningful engagement. However, understanding students' behavioural intentions alone is insufficient. Attention should also shift towards the quality of students' engagement with GenAI to evaluate whether they use it critically, reflectively and purposefully rather than passively and dependently. Appropriate reliance and engagement with GenAI can enhance learning efficiency, but excessive reliance or uncritical usage can make students' thinking and skill development suffer. By investigating students' intention, critical use, and reliance on GenAI, educators and institutions can design strategies that maximise GenAI's pedagogical benefits while mitigating its risks.

Recent reviews of GenAI use in higher education focus largely on technology acceptance (Mukhamedkarimova & Umurkulova, 2025; Park, 2025), overlooking the factors that shape students' reliance on GenAI. This gap is notable because emerging evidence demonstrates substantial variation in how heavily students depend on GenAI and how critically they evaluate its outputs (Stojanov et al., 2024). These findings suggest that reliance is a distinct behavioural construct influenced by cognitive and evaluative processes, yet the role of students' critical technology use in shaping reliance remains underexplored.

This oversight is particularly consequential in disciplines where accuracy and analytical problem-solving are essential. Despite being early, frequent and technically capable adopters of GenAI tools, engineering and CS/IT students remain significantly underrepresented in the current literature. Although these students report higher GenAI knowledge and stronger intentions to use such tools than peers in arts, humanities and social sciences (Qu et al., 2024), little is known about the quality of their engagement. Accordingly, this study aims to investigate factors influencing intention to use and reliance on GenAI among engineering and IT students, with an emphasis on critical use.

## 2. Theoretical background and hypothesis development

### 2.1. Frameworks for examining GenAI adoption

The adoption of GenAI in higher education can be examined through established technology adoption frameworks that explain how individuals accept and integrate new technologies. Theoretical models provide a foundation for explaining the relationships between key constructs and predicting behavioural outcomes (Green et al., 2007; as cited in Hsiao & Tang, 2025). Given this consideration, most existing work has drawn on the Technology Acceptance Model (TAM) (Li et al., 2024; Shen et al., 2025) or the Unified Theory of Acceptance and Use of Technology (UTAUT) (Venkatesh et al., 2003; Zhao et al., 2025) to explain learners' engagement with newly-introduced technology within the education sector.

Owing to its simplicity and predictive validity, TAM, proposed by Davis (1989), has been widely applied to examine technology adoption across various contexts, including e-learning systems, mobile platforms, and AI-driven learning tools (Ma, 2025). Grounded in information systems theory, TAM proposes that two core factors, including Perceived Usefulness (PU) and Perceived Ease of Use (PEU), shape users' Attitude towards using the technology (ATT). ATT then predicts Behavioural Intention (BI) to use the technology, which ultimately influences actual usage behaviour (Davis, 1989, 1993). TAM serves as a strong foundation to understand behavioural patterns related to GenAI adoption. TAM

foregrounds cognitive and psychological factors that drive users to utilise new technologies (Granić & Marangunić, 2019), making it suitable for examining individual engagement with GenAI tools in higher education. However, as AI technologies become increasingly autonomous, interactive, and contextually complex, a question arises about whether traditional TAM sufficiently captures adoption intentions and behavioural patterns, such as the depth of cognitive engagement. TAM, therefore, has been extended and refined to capture a broader range of psychological and socio-cultural determinants of technology adoption (Venkatesh & Bala, 2008; Venkatesh & Davis, 2000).

Across recent GenAI adoption studies – conducted in China (Li et al., 2024; Liu & Ma, 2024; Shen et al., 2025), Hong Kong (Lai et al., 2023), Thailand (Kanont et al., 2024), Pakistan (Shahzad et al., 2025), Oman (Tiwari et al., 2024), and Bangladesh (Rahman et al., 2023) – a consistent pattern emerges regarding the application of TAM. Most studies have either extended TAM with additional predictors or integrated it with other theoretical frameworks, such as Self-Determination Theory in Lai et al. (2023) and Shen et al. (2025). This trend suggests that the original TAM alone is increasingly viewed as insufficient for capturing the complexity of students' acceptance and use of GenAI tools. Shen et al. (2025), for instance, integrated TAM with Self-Determination Theory to explain art and design students' adoption of GenAI-image tools, depicting deeper motivational, emotional, and ethical issues related to originality, privacy and job displacement. Within higher education, Kanont et al. (2024)'s study demonstrated a surprising negative relationship between PEU and PU. This suggests that GenAI challenges the predictive logic of TAM by introducing new behavioural dynamics beyond traditional PEU, PU and ATT. Building on this emerging pattern of extending TAM in GenAI adoption research, the present study adopts and extends TAM by introducing self-reported critical use as a theoretically and practically important construct. This extension is based on the premise that simply introducing technology in higher education is insufficient to ensure its effective application (Mei et al., 2018), that technology acceptance alone is insufficient to fully capture post-adoption behavioural patterns, and that acceptance alone provides no assurance of critical use. Moreover, while students' ability to critically evaluate information has been shown to influence their engagement with educational technologies (Zhao, 2024), its potential influence on GenAI adoption remains underexplored.

Although UTAUT offers a comprehensive framework by incorporating social and organisational determinants (Venkatesh et al., 2003), TAM was considered more appropriate for this study for two reasons. First, TAM has fewer parameters, which facilitates clearer theoretical extension and minimises potential confounding effects arising from complex interrelationships among multiple predictors. As this study seeks to examine students' critical use and reliance on GenAI, employing a simpler base model enables a more precise assessment of these added constructs. Second, the data were collected from Engineering and CS/IT students within a single institution, representing a relatively homogeneous academic context. Meanwhile, UTAUT places strong emphasis on social influences, which are typically more salient in multi-organisational environments. Given the bounded educational setting of this study, TAM is considered more conceptually aligned and methodologically suitable.

It is also noteworthy that previous research on TAM and GenAI has focused on China and other developing countries, while studies from other contexts, including Australia, are scarce. Additionally, very few studies (Prather et al., 2023; Santos et al., 2024; Wilson & Nishimoto, 2024) have explicitly examined GenAI use among Engineering, Computer Science or Information Technology (CS/IT) students, despite their central and increasing engagement with AI-driven technologies. Science, Technology, Engineering and Mathematics (STEM) students are generally more inclined to adopt digital tools for coding, debugging, and technical problem-solving than non-STEM students, who tend to use such tools primarily for writing or instructor-directed tasks (Arum et al., 2025). However, the assumption that GenAI is highly effective for

technical applications remains questionable. ChatGPT, in fact, struggles with complex programming tasks, hence its inconsistent or inaccurate outputs (Qureshi, 2023; as cited in Kosar et al., 2024). Kosar et al. (2024) found that ChatGPT demonstrated limitations in Computer Engineering contexts, particularly in code completion and tasks involving visual inputs. Moreover, while CS/IT and Engineering students' technical proficiency may facilitate GenAI adoption, it is not necessarily associated with their intention to use or critical usage. These gaps underscore the need for context-specific research to extend current understanding of GenAI adoption among Engineering and CS/IT students in higher education.

## 2.2. Students' critical use and reliance on GenAI

Although TAM and its extensions offer valuable insights into the drivers of GenAI adoption, there is a lack of in-depth understanding of critical use and reliance on GenAI. Specifically, existing research has frequently drawn on related terms such as information literacy and digital competence to discuss students' preparedness for using technologies (Jose et al., 2024; Pinto et al., 2021; Sarva et al., 2023; Zhao et al., 2021). However, AI literacy and critical use of GenAI are emerging terms associated with the recent rise of GenAI tools, and reliance on GenAI remains even less systematically investigated within higher education contexts (Stojanov et al., 2024). Meaningful technology adoption requires more than exposure; it depends on the ability to question, evaluate, and responsibly apply digital tools. In this regard, AI literacy has become a foundational competence, encompassing the understanding of human-AI interaction, knowledge of how AI processes data, and practical experience with AI applications (Walter, 2024). Often described as essential as reading and writing in the twenty-first century (Zhang et al., 2023), AI literacy enables learners to navigate GenAI's strengths and limitations. Similarly, critical use of GenAI involves recognising its affordances, interrogating the reliability of its outputs, and refining responses through human judgement (Michel-Villareal et al., 2023; Stojanov et al., 2024). Students with stronger critical thinking and disciplinary knowledge tend to engage with GenAI as partners in inquiry rather than shortcuts to answers (Rahimi, 2025). Stojanov et al. (2024) further show that students who rely more heavily on ChatGPT also demonstrate higher levels of critical evaluation, suggesting a meaningful link between critical use and responsible reliance.

Reliance on GenAI is influenced by multiple factors, such as perceived usefulness, ease of use, trust, AI literacy, task complexity, assessment demands, and institutional norms (Stojanov et al., 2024; Zhai et al., 2024). Liu et al. (2025) find that performance and effort expectancy enhance GenAI usage intention, whereas perceived threats and low AI literacy reduce it. This aligns with broader research indicating that AI literacy underpins critical use, as learners who understand how GenAI generates, filters and occasionally fabricates information are better equipped to question the validity and applicability of its outputs (Bećirović et al., 2025; Veldhuis et al., 2025).

Importantly, reliance that exceeds reflective evaluation falls short of what is understood as critical use of GenAI. One clear manifestation of over-reliance is the uncritical acceptance of AI-generated outputs without any form of verification (Gao et al., 2022). Such excessive dependence can promote shallow engagement, increase exposure to misinformation, and heighten the likelihood of academic misconduct (Dwivedi et al., 2023; Jacovi et al., 2021). Students may gradually outsource judgment to the system, reducing their capacity to make independent, well-informed decisions (Bućinca et al., 2021). Tian and Zhang (2025) also added that over-reliance on AI without critical evaluation results in reduced deep learning engagement and weaker critical analysis. These risks are amplified by GenAI's known limitations, including biased, inconsistent, and factually incorrect outputs (Stojanov, 2023). As a result, over-reliance raises concerns regarding AI hallucinations, transparency, plagiarism, and data privacy, all of which threaten both learning outcomes and institutional integrity (Abbas et al.,

2025; Bjelobaba et al., 2025; Nguyen, 2025; Zhai et al., 2024). Despite the seriousness of these issues, research examining the predictors of students' reliance on GenAI remains scarce. The extent to which students depend on GenAI for different types of tasks is also underexplored, and little is known about how this reliance unfolds in more specialised disciplinary activities. At the same time, existing studies on other educational technologies have shown that users' critical digital capacities, such as digital competence and information literacy, significantly influence their actual use of digital tools and online resources for academic learning (Zakir et al., 2025; Zhao, 2024). Nevertheless, research investigating how critical use shapes the adoption and reliance of GenAI tools remains limited.

## 2.3. Hypothesis development

As discussed above, while TAM and its extensions offer valuable insights into the fundamental factors shaping GenAI adoption, they fall short in explaining students' intention to use and reliance on GenAI for academic purposes. Accordingly, this study examines the factors influencing GenAI intention and reliance among Engineering and CS/IT students by extending TAM with the construct of self-reported critical use. Reliance on GenAI is conceptualised across four functional domains: understanding, assessment, programming and engineering project tasks. The first two domains (understanding and assessment) represent core academic activities across disciplines. Reliance on GenAI for understanding includes using the tool to explain concepts, clarify ideas, or summarise learning materials, which supports sense-making during the learning process. Reliance in assessment contexts reflects the use of GenAI to prepare assignments, draft responses, or organise academic work. The remaining two domains (programming and engineering project tasks) are specific to Engineering and CS/IT education. Programming-related tasks involve activities such as code generation, debugging, and optimisation, all of which require technical precision and domain expertise. Engineering projects involve higher-order tasks such as problem formulation, system design, modelling, and iterative decision-making, which are central to engineering practice and professional competence. The inclusion of general and discipline-specific domains, therefore, enables a more insightful investigation of students' GenAI adoption.

A conceptual framework (Fig. 1) is introduced to guide the investigation. Perceived ease of use (PEU) influences perceived usefulness (PU) and attitude (ATT) towards technology adoption. When a technology is perceived as easy to use, it leads to a greater sense of usefulness and a more positive attitude towards the tool (Mailizar et al., 2021; Rispler et al., 2025). In addition, PEU is proposed to influence critical use (CU), where users engage with the tool in a more thoughtful and responsible manner. Students who find GenAI tools easier to use are more likely to engage in critical use, verifying and assessing the generated content. Beyond these primary relationships, the model incorporates additional factors like student status, CU, intention (INT), and reliance (RE), which play key roles in explaining GenAI adoption and its use in higher education. Student status accounts for variation in enrolment types, which may shape how students engage with GenAI. INT reflects students' tendency to adopt GenAI, but intention alone does not fully capture the depth of engagement with GenAI tools. Therefore, reliance across four functional domains is introduced to move beyond simple adoption and reflect the degree of dependence in different academic tasks. These relationships are central to the hypotheses below.

**H1.** Perceived Ease of Use (PEU), Perceived Usefulness (PU), and Attitude (ATT) positively influence intention to use GenAI (INT).

**H2.** Critical Use (CU) positively influences intention to use GenAI (INT).

**H3.** Intention to use GenAI (INT) positively predicts reliance (RE) across four functional domains: understanding, assessment,

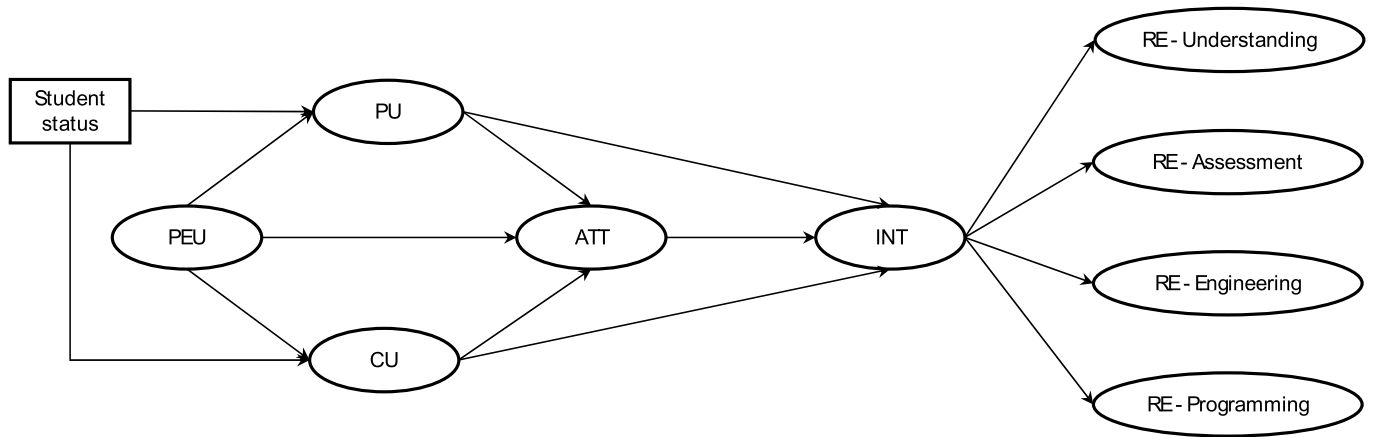


Fig. 1. Conceptual framework.

programming, and engineering projects.

### 3. Methods

#### 3.1. Participants

An anonymous online survey was administered via the QuestionPro platform between April and June 2025. It was advertised through posts on student online hubs and flyers on noticeboards around the campus. Inclusion criteria were being at least 18 years old and currently enrolled as an Engineering or CS/IT student. Ethics approval was obtained from the University (HEC25057). The survey took approximately 15 min. A total of 126 students completed the survey, primarily from Engineering (65.1%) and Computer Science/IT (34.9%) programs. Almost half of the participants were aged 18-24 years (48.4%), followed by 25-30 years (40.5%), and a smaller proportion over 30 years (11.1%). The majority were male (69.1%) and international (65.9%). Regarding education levels, about 54% were postgraduate students, and 46% were undergraduate students. A summary of the sample characteristics is presented in Table 1.

#### 3.2. Procedure and measures

After reviewing the participant information statement and consent form and indicating their consent, participants completed several demographic questions, as detailed in Table 1. Then, they completed survey questions measuring PEU, PU, ATT, CU, INT and RE (for understanding, assessment, programming and engineering projects). A summary of the questions is presented in Table 2.

Using a 7-point Likert scale (1 = strongly disagree and 7 = strongly agree), the PEU, PU, ATT and INT constructs were measured in alignment with prior TAM-based ChatGPT acceptance studies (Bernabei

Table 1

Sample characteristics (n = 126).

| Variable        | Category            | Count | Percent |
|-----------------|---------------------|-------|---------|
| Age group       | 18-24               | 61    | 48.41%  |
|                 | 25-30               | 51    | 40.48%  |
|                 | >30                 | 14    | 11.11%  |
| Gender          | Female              | 38    | 30.16%  |
|                 | Male                | 87    | 69.05%  |
|                 | Non-binary          | 1     | 0.79%   |
| Student status  | Domestic            | 43    | 34.13%  |
|                 | International       | 83    | 65.87%  |
| Education level | Undergraduate       | 58    | 46.03%  |
|                 | Postgraduate        | 68    | 53.97%  |
| Study field     | Engineering         | 82    | 65.08%  |
|                 | Computer Science/IT | 44    | 34.92%  |

Table 2

Summary of the Perceived Ease of Use, Perceived Usefulness, Attitudes, Critical Use, and Intention to Use GenAI constructs and measuring items.

| Constructs and items                                                                              | Mean  | SD    |
|---------------------------------------------------------------------------------------------------|-------|-------|
| Perceived Ease of Use - PEU ( $\alpha = 0.799$ , AVE = 0.577)                                     |       |       |
| It is easy to learn how to use GenAI tools like ChatGPT                                           | 5.444 | 1.354 |
| I find GenAI tools simple to navigate and use for academic purposes                               | 5.246 | 1.407 |
| The instructions and interface of GenAI tools are user-friendly                                   | 5.540 | 1.244 |
| Overall                                                                                           | 5.410 | 1.128 |
| Perceived Usefulness - PU ( $\alpha = 0.871$ , AVE = 0.631)                                       |       |       |
| I find GenAI tools (e.g., ChatGPT) helpful for enhancing my academic performance                  | 5.214 | 1.484 |
| GenAI tools save me time in completing academic tasks                                             | 5.111 | 1.645 |
| Using GenAI tools improves the quality of my academic work                                        | 4.937 | 1.648 |
| GenAI tools help me understand complex topics better                                              | 5.540 | 1.537 |
| Overall                                                                                           | 5.200 | 1.341 |
| Attitudes - ATT ( $\alpha = 0.912$ , AVE = 0.721)                                                 |       |       |
| I enjoy using GenAI tools (e.g., ChatGPT) for academic purposes                                   | 4.802 | 1.644 |
| GenAI tools make my study process more engaging                                                   | 5.008 | 1.612 |
| I feel comfortable integrating GenAI tools into my daily academic activities                      | 4.778 | 1.846 |
| I have a positive attitude towards using GenAI for education purposes                             | 5.111 | 1.641 |
| Overall                                                                                           | 4.925 | 1.499 |
| Critical Use - CU ( $\alpha = 0.809$ , AVE = 0.593)                                               |       |       |
| I always verify information provided by GenAI with other trusted sources                          | 5.611 | 1.453 |
| I use responses provided by GenAI as a starting point for assessments, not the final answer       | 5.278 | 1.787 |
| I approach GenAI's responses with a critical mindset, questioning the information it provides     | 5.262 | 1.626 |
| Overall                                                                                           | 5.384 | 1.382 |
| Intention to Use GenAI - INT ( $\alpha = 0.871$ , AVE = 0.700)                                    |       |       |
| I intend to use GenAI tools (e.g., ChatGPT) for academic purposes in the future                   | 4.667 | 1.789 |
| I am likely to explore new GenAI tools to assist in my learning activities                        | 4.810 | 1.672 |
| I expect to use GenAI tools for personal learning and skill development beyond my current studies | 5.230 | 1.601 |
| Overall                                                                                           | 4.902 | 1.506 |

et al., 2023; Liu & Ma, 2024). The PEU construct includes three items that assess participants' perceived effort in learning and operating GenAI for academic use. Four items were used in the PU construct for evaluating participants' perceived practical and performance-related benefits of using GenAI. The ATT construct was measured using four items capturing participants' overall attitudes towards employing GenAI in their studies, including enjoyment, engagement, comfort, and positivity. Finally, the INT construct was measured using three items evaluating participants' future behavioural intentions and willingness to engage with GenAI tools.



The CU construct was measured using three items assessing participants' ability to engage reflectively and responsibly with GenAI tools by verifying and critically evaluating generated content. Although some AI literacy instruments have been developed, such as a 12-item scale covering awareness, usage, evaluation, and ethics (Wang et al., 2023), scales focusing explicitly on critical use remain limited. Accordingly, the three CU items in this study were developed through adaptation and refinement of relevant items embedded within these broader AI literacy scales and questionnaires (Carolus et al., 2023; Ng et al., 2024; Wang et al., 2023) and critical use of ChatGPT (Stojanov et al., 2024). The selected items were aligned conceptually with the definition of critical use as reflective evaluation and responsible application of AI-generated content. The wording of the items was adjusted to suit the higher education context and GenAI usage in academic tasks. Participants rated their agreement with statements using a 7-point Likert scale (1 = strongly disagree and 7 = strongly agree). These items align with prior research on ChatGPT usage and adoption (Shahzad et al., 2024). A 7-point Likert scale was selected because it is widely used in survey research (Weijters et al., 2010), and offers greater sensitivity, variability, and measurement reliability compared to shorter response formats (Alwin, 1997).

The RE constructs, partly derived from previous GenAI reliance research (Stojanov et al., 2024), were measured across four functional dimensions (i.e., understanding, assessment, programming, and engineering projects) to capture the extent to which students depend on GenAI for different academic purposes (see Table 3). A 7-point Likert scale (1 = not at all reliant and 7 = completely reliant) was adopted. The understanding dimension assessed how students use GenAI to clarify complex concepts, summarise texts, or find answers to study-related questions. The assessment dimension focused on the use of GenAI for drafting, editing, and obtaining feedback on assessments. The programming dimension measured students' reliance on GenAI for code generation and debugging, while the engineering projects dimension examined its use in performing technical calculations and generating design ideas. These dimensions align with prior findings that students employ ChatGPT for diverse cognitive and practical learning tasks, ranging from comprehension and writing support to computational and design applications.

### 3.3. Data analysis

Partial Least Squares Structural Equation Modelling (PLS-SEM) was

**Table 3**  
Summary of the Reliance on GenAI construct and measuring items.

| Reliance on GenAI (RE)                                                                   | Mean  | SD    |
|------------------------------------------------------------------------------------------|-------|-------|
| RE - Understanding ( $\alpha = 0.787$ , AVE = 0.557)                                     |       |       |
| Find answers to specific questions related to studies                                    | 3.683 | 1.573 |
| Understand complex concepts                                                              | 4.849 | 1.585 |
| Summarise complex texts                                                                  | 4.762 | 1.531 |
| Overall                                                                                  | 4.431 | 1.309 |
| RE - Assessment ( $\alpha = 0.876$ , AVE = 0.587)                                        |       |       |
| Draft outlines of written assessments                                                    | 3.976 | 1.874 |
| Help with writing assignments like suggesting words, referencing, or proofreading        | 3.841 | 1.857 |
| Finding or summarising scholarly articles for assessments                                | 3.921 | 1.814 |
| Obtain feedback on written assignments                                                   | 3.897 | 1.930 |
| Write assessments entirely                                                               | 2.825 | 1.838 |
| Overall                                                                                  | 3.692 | 1.522 |
| RE - Programming ( $\alpha = 0.883$ , AVE = 0.794)                                       |       |       |
| Complete coding or programming tasks                                                     | 3.794 | 1.920 |
| Identify and fix bugs in programming code                                                | 3.841 | 1.920 |
| Overall                                                                                  | 3.817 | 1.816 |
| RE - Engineering Projects ( $\alpha = 0.857$ , AVE = 0.751)                              |       |       |
| Perform calculations or provide formulas for engineering design processes                | 3.556 | 1.827 |
| Generate initial design concepts for engineering projects or ideas for creative projects | 3.611 | 1.829 |
| Overall                                                                                  | 3.583 | 1.710 |

employed to examine the hypothesised relationships among constructs. As a variance-based structural modelling approach, PLS-SEM is particularly suitable for analysing complex models with multiple latent variables and relatively small to medium sample sizes. This approach has been applied in recent studies investigating ChatGPT adoption in higher education (Lai et al., 2023). In this study, PLS-SEM analyses were performed using the *SEMinR* package in R (Hair & Alamer, 2022). Latent constructs (PEU, PU, CU, ATT, INT, and RE) were specified as reflective measurement models based on the items presented in Tables 2 and 3. Student status was incorporated as a single-indicator control variable. Internal consistency reliability and convergent validity were assessed through Cronbach's  $\alpha$  and Average Variance Extracted (AVE), respectively, while multicollinearity was evaluated using the Variance Inflation Factor (VIF). The Heterotrait–Monotrait (HTMT) ratio was employed to assess discriminant validity. To determine the significance of path coefficients, a bootstrapping procedure with 5000 samples was applied.

## 4. Results

### 4.1. Descriptive statistics

A summary of the key constructs is presented in Table 2. Overall, students reported moderately high perceptions of both ease of use (mean = 5.41) and usefulness (mean = 5.20), indicating that GenAI tools such as ChatGPT were generally viewed as accessible and beneficial for academic tasks. Within ease of use, the highest-rated item was about user-friendliness (mean = 5.54), suggesting strong usability perceptions. Perceived usefulness was reinforced by students' agreement that GenAI helps them understand complex topics better (mean = 5.54). Attitudes towards GenAI were also positive (mean = 4.93), reflecting enjoyment and comfort in integrating these tools into learning activities. Importantly, participants demonstrated a high level of critical use (mean = 5.38), showing awareness of the need to verify GenAI outputs and use them as a starting point rather than a final answer. Finally, intention to use (mean = 4.90) was moderately strong, with respondents expressing interest in continuing to explore and apply GenAI for learning purposes. Reliability and validity statistics (Cronbach's  $\alpha = 0.799$  to 0.912; AVE = 0.577 to 0.721) confirmed robust measurement reliability and validity across all constructs.

Table 3 presents the descriptive statistics for students' reliance on GenAI tools across four dimensions: understanding, assessment, programming, and engineering projects. The internal consistency and convergent validity of all constructs were strong (Cronbach's  $\alpha = 0.787$  to 0.883; AVE = 0.557 to 0.794). Overall, students reported a moderate level of reliance on GenAI, with the highest mean observed for understanding-related activities (mean = 4.43). This suggests that students predominantly use GenAI as a useful tool to clarify difficult concepts, summarise complex texts, and find answers to study-related questions, reflecting its growing role as a digital learning assistant rather than a content generator. Reliance on assessment-related tasks was lower (mean = 3.69), indicating selective use of GenAI for generating outlines, proofreading, and summarising scholarly content, but limited reliance on it for writing full assessments (mean = 2.83). This suggests cautious and critical engagement, reflecting students' awareness of ethical and academic integrity issues when using GenAI for written work. Reliance on GenAI for programming tasks (mean = 3.82) and engineering projects (mean = 3.58) was moderate, indicating that students value GenAI's assistance for debugging, formula generation, and design ideation, while still maintaining an active role in problem-solving. These patterns indicate that engineering and computer-science students perceive GenAI as a supplementary problem-solving tool rather than a substitute for technical competence.

In addition, HTMT values for all constructs were below the threshold of 0.90, indicating adequate discriminant validity. This finding confirms that the constructs were distinct, particularly with respect to RE across

the four domains.

#### 4.2. Results of the PLS-SEM model

The estimated results of the PLS-SEM model are presented in Fig. 2. The R-squared for INT was 0.60, indicating a strong explanatory power. In other words, the model explained about 60% of the variance in INT. The VIF values were below 4, indicating no evidence of multicollinearity. All indicator loadings were above 0.6 and statistically significant, which confirmed the robustness of the indicators in forming the constructs.

PEU had a strong positive effect on PU (path coefficient ( $\beta$ ) = 0.7, 95% confidence interval (CI): 0.54 to 0.84), which in turn had a strong positive effect on ATT ( $\beta$  = 0.88, 95% CI: 0.66 to 1.15). As expected, ATT also had a strong positive effect on INT ( $\beta$  = 0.64, 95% CI: 0.10 to 1.20). However, the paths between PEU and ATT and between PU and INT were insignificant. The results generally show that the TAM constructs, including PEU, PU, and ATT, had direct or indirect effects on INT, confirming the suitability of TAM for predicting intention to use GenAI. Total effects, including direct and any indirect effects, are further presented in Table 4. The total effects of PEU, PU and ATT on INT were 0.49 (95% CI: 0.31 to 0.65), 0.58 (95% CI: 0.38 to 0.82), and 0.64 (95% CI: 0.10 to 1.20), respectively. These results supported hypothesis H1, which predicted that PEU, PU, and ATT would have direct or indirect positive effects on INT, in accordance with TAM.

It was also evident that PEU had a moderate positive effect on CU ( $\beta$  = 0.34, 95% CI: 0.09 to 0.58). While CU was not significantly associated with ATT, it had a modest positive effect on INT ( $\beta$  = 0.27, 95% CI: 0.10 to 0.46). This finding supported hypothesis H2, which posited that CU would have a positive effect on INT.

Fig. 2 shows that INT had a strong positive effect on reliance for understanding ( $\beta$  = 0.74, 95% CI: 0.60 to 0.86), followed by reliance for programming ( $\beta$  = 0.60, 95% CI: 0.44 to 0.73) and reliance for assessment tasks ( $\beta$  = 0.56, 95% CI: 0.40 to 0.70). INT had a moderately positive effect on reliance for engineering projects ( $\beta$  = 0.38, 95% CI: 0.17 to 0.56). Thus, hypothesis H3 was also supported.

Student status had a significant positive relationship with PU ( $\beta$  = 0.22, 95% CI: 0.06 to 0.37). Specifically, international students tend to have higher PU than domestic students. It is worth noting that predicting INT with the three original TAM constructs (PEU, PU and ATT) and student status explained 53.5% of the variance in INT (R-squared = 0.54). Thus, extending the TAM model with CU for predicting INT significantly improved its explanatory power (R-squared = 0.60).

Table 4 also shows the positive effects of PEU, PU, ATT, CU, and student status on students' reliance on GenAI for the four activities. Similar to the effects on INT, international students tend to rely more on

**Table 4**

Total effects on Intention to Use GenAI and Reliance on GenAI.

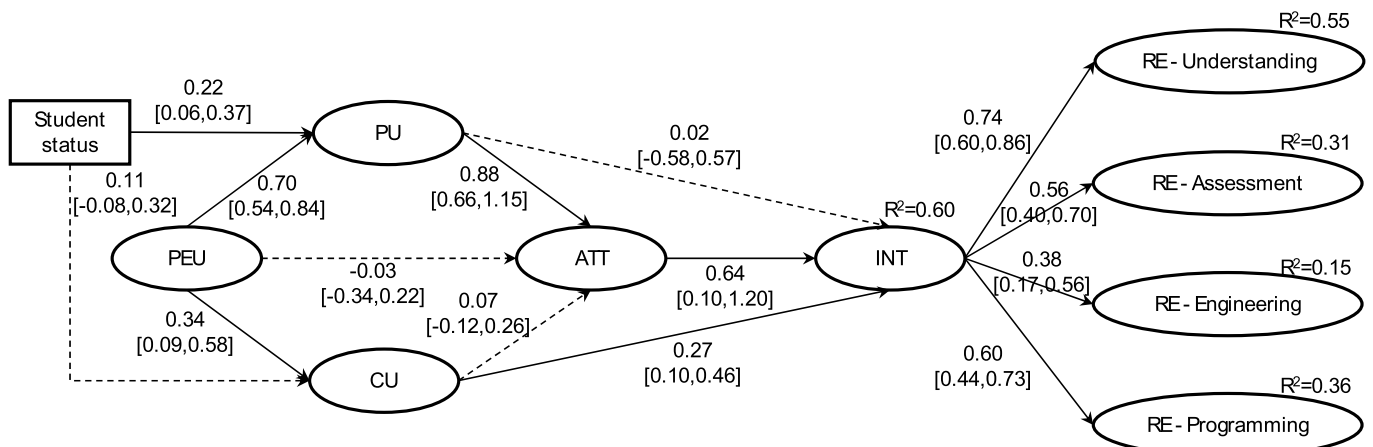
| Paths                                      | Total effects | 2.5% CI | 97.5% CI |
|--------------------------------------------|---------------|---------|----------|
| PEU → INT                                  | 0.493         | 0.309   | 0.654    |
| PEU → RE - Assessment                      | 0.275         | 0.140   | 0.422    |
| PEU → RE - Understanding                   | 0.364         | 0.203   | 0.531    |
| PEU → RE - Programming                     | 0.294         | 0.160   | 0.435    |
| PEU → RE - Engineering Projects            | 0.188         | 0.068   | 0.322    |
| PU → INT                                   | 0.582         | 0.380   | 0.815    |
| PU → RE - Assessment                       | 0.325         | 0.171   | 0.516    |
| PU → RE - Understanding                    | 0.430         | 0.254   | 0.654    |
| PU → RE - Programming                      | 0.348         | 0.206   | 0.513    |
| PU → RE - Engineering Projects             | 0.222         | 0.086   | 0.383    |
| ATT → INT                                  | 0.644         | 0.099   | 1.201    |
| ATT → RE - Assessment                      | 0.360         | 0.050   | 0.715    |
| ATT → RE - Understanding                   | 0.476         | 0.068   | 0.935    |
| ATT → RE - Programming                     | 0.385         | 0.057   | 0.725    |
| ATT → RE - Engineering Projects            | 0.245         | 0.024   | 0.533    |
| CU → INT                                   | 0.312         | 0.096   | 0.520    |
| CU → RE - Assessment                       | 0.175         | 0.051   | 0.316    |
| CU → RE - Understanding                    | 0.231         | 0.069   | 0.400    |
| CU → RE - Programming                      | 0.187         | 0.054   | 0.337    |
| CU → RE - Engineering Projects             | 0.119         | 0.024   | 0.248    |
| Student Status → INT                       | 0.163         | 0.040   | 0.289    |
| Student Status → RE - Assessment           | 0.091         | 0.022   | 0.175    |
| Student Status → RE - Understanding        | 0.120         | 0.029   | 0.221    |
| Student Status → RE - Programming          | 0.097         | 0.024   | 0.177    |
| Student Status → RE - Engineering Projects | 0.062         | 0.012   | 0.130    |

GenAI for the four activities than domestic students do. The results show that CU had modest effects on reliance for understanding (0.23, 95% CI: 0.07 to 0.40), followed by reliance for programming (0.19, 95% CI: 0.05 to 0.34), reliance for assessment tasks (0.18, 95% CI: 0.05 to 0.32), and engineering projects (0.12, 95% CI: 0.02 to 0.25). These findings indicate that students who approach GenAI outputs with a critical and evaluative mindset are more likely to integrate such tools thoughtfully across various academic applications. When compared to other predictors (including PEU, PU and ATT), CU's influence was smaller. These results suggest that while ease of use, usefulness and positive attitudes remain the primary drivers of GenAI adoption, critical use plays a distinct, complementary role by shaping more intentional and responsible engagement with GenAI-assisted learning tools.

## 5. Discussion

### 5.1. Key findings

The present research utilised an extended TAM by incorporating self-reported critical use and by conceptualising reliance on GenAI across four functional domains. All three hypotheses were supported,



**Fig. 2.** PLS-SEM results. Note: 95% confidence intervals (CIs) are shown in brackets; dashed lines show insignificant path coefficients.

demonstrating that these extensions enhance the model's explanatory power for understanding university students' intention to use and reliance on GenAI. Notably, the findings related to H2 and H3 represent novel contributions to the emerging GenAI literature.

The results of H1 indicate that PEU, PU, and ATT exert direct or indirect positive effects on INT. Specifically, PEU strongly influenced PU, which then had a pronounced positive effect on ATT, and ATT emerged as a strong predictor of INT. These patterns mirror Stroud and Du (2025)'s findings on ChatGPT acceptance in higher education. Moreover, the significant link from PEU to PU also echoes results reported in prior TAM studies (Alfadda & Mahdi, 2021; Liu & Ma, 2024). This relationship reinforces the idea that the simplicity and convenience of GenAI tools encourage students to view them as beneficial for learning. Also noteworthy is the insignificant direct effect of PEU on ATT, a result consistent with Liu and Ma (2024) but contrary to findings from Bailey et al. (2022) and Supriyanto et al. (2024). This suggests that while students acknowledge the usability and accessibility of GenAI tools, ease of use alone does not foster positive attitudes towards GenAI. In contrast, Bailey et al. (2022) reported a positive effect of PEU on attitudes towards technology, specifically in the context of video-conferencing tools, while Supriyanto et al. (2024) found that PEU significantly influenced attitudes towards using ChatGPT. Additionally, the non-significant direct effect of PU on INT indicates that even when engineering and IT students recognise GenAI's value, this recognition does not automatically translate into intention to use.

It is also worth noting that PU and PEU may be influenced by GenAI's design to respond in ways that appeal to the users. Since ChatGPT has the mentalising ability to comprehend and interpret users' mental states, such as their thoughts and intentions (Hadar-Shoval et al., 2023), this impact is plausible. Furthermore, GenAI tools are built to tailor responses to users' queries and adjust to subtle cues in language and behaviour (Abels et al., 2025; Fang & Zhou, 2025). This may have an impact on users' PU and PEU. In engineering education, GenAI tools have been proven to offer benefits such as enhancing learning efficiency, supporting creativity, and aiding in understanding challenging concepts (Fan et al., 2025). However, limitations must also be acknowledged, as these tools can produce inaccurate content, raise concerns over data privacy and copyright, and struggle with more specialised engineering problems (Fan et al., 2025; Munir et al., 2025). For tasks like general engineering design and concept explanations, GenAI can be useful. However, for tasks requiring exact mathematical accuracy or complicated computational analysis, the tools often fall short (Munir et al., 2025). Moreover, the effectiveness of prompting strategies is a crucial factor in shaping user perceptions. Hwang et al. (2025) demonstrated that students who tend to use shorter, concise and well-defined prompts have higher satisfaction. Interestingly, prompts in students' native language often resulted in lower satisfaction than those in English (Hwang et al., 2025).

H2 suggests that CU would have a positive effect on INT. This indicates that students who actively evaluate, question, and cross-check GenAI outputs are more motivated to integrate the tools into their learning. This relationship is plausible in light of documented student concerns about GenAI. Stroud and Du (2025), for instance, found that some students fear that frequent use of ChatGPT may reduce opportunities for independent reasoning and problem-solving. Because CU involves skills such as validating information and triangulating AI-generated content, students who feel unprepared or lack confidence in these practices may be hesitant to adopt GenAI tools more broadly. Although similar patterns have been reported in research on other educational technologies, where digital competence and information literacy positively impact technology acceptance and digital engagement (Scheel et al., 2022; Zhao, 2024), the existing literature has not explicitly demonstrated a link between CU and INT as shown in this study. The findings also show that incorporating CU into TAM substantially enhanced the model's ability to predict INT.

H3 posits that INT positively predicts RE across four functional

domains: understanding, assessment, programming, and engineering projects, and the results support this hypothesis. This finding indicates that engineering and IT students' reliance on GenAI is strongly grounded in their intention to use the technology for these distinct academic purposes. Although research on this relationship is scant, existing studies (Alfadda & Mahdi, 2021; Liu & Ma, 2024; Sun & Mei, 2022) have shown that behavioural intention predicts actual use of educational technologies. At the same time, contradictory evidence reminds us that this relationship is not always linear. Ruyobeza et al. (2025) reported that behavioural intention does not always translate to actual usage. Their analysis notes the influence of other factors such as contextual constraints or situational barriers, which can weaken the link between behavioural intention and actual use.

Additionally, previous studies have described how students use GenAI for general academic purposes, such as improving understanding or clarifying concepts, as well as for specialised tasks like coding (Hazzan & Erez, 2025). However, the present study advances the literature by documenting a clear positive link between intention to use and reliance across multiple functional domains. This offers an important contribution to engineering and CS/IT education, where the alignment between GenAI's functionality and the cognitive, computational, and design-oriented demands of coursework may make the intention-reliance connection especially compelling.

Among the three forms of reliance, including over-reliance, appropriate reliance, and under-reliance as discussed in Schemmer et al. (2023), the participants in this study demonstrated appropriate reliance with a moderate level of dependence on GenAI. Students reported relying most heavily on GenAI for understanding-related activities, such as clarifying concepts or obtaining worked examples, and least for writing full assessments, reflecting a cautious, regulated use aligned with academic integrity expectations. This tendency to rely on GenAI for understanding-related tasks is consistent with previous findings (Fang & Zhou, 2025; Hazzan & Erez, 2025; Lee et al., 2025; Maxwell et al., 2025). This pattern is justifiable given that GenAI tools are known to be effective at providing instant explanations and breaking down abstract concepts into simpler terms (Perifanou & Economides, 2025). Such affordances align closely with cognitive support functions, enabling students to scaffold their learning, address knowledge gaps, and access immediate feedback (Fang & Zhou, 2025). Reliance on GenAI for understanding-related tasks, rather than high-stakes assessments, may also reflect students' awareness of the technology's limitations. Such limitations are particularly problematic in disciplines like engineering, which are highly technical and require precision and up-to-date knowledge (Fan et al., 2025). Consequently, students may be less willing to rely on GenAI for graded academic work. Dependence on GenAI tools for programming tasks and engineering project work was also moderate. This suggests that while students draw on GenAI for debugging, formula generation, and design ideation, they continue to maintain an active role in analytical reasoning and problem-solving. Such selective, purposeful engagement demonstrates students' understanding of both the capabilities and limitations of GenAI. They recognise that although GenAI has certain educational benefits, it cannot replace the deeper disciplinary thinking required for accurate, context-sensitive engineering and computer science tasks. This discernment is a key indicator of functional AI literacy, which encompasses not only the technical skills to interact with AI but also the judgement required to evaluate AI outputs and integrate them meaningfully into their own work.

Finally, these findings suggest that students who engage critically with GenAI outputs are more likely to use the tool thoughtfully across academic tasks. Although its effect was smaller than that of PEU, PU and ATT, critical use still contributes meaningfully by supporting more intentional and responsible engagement with GenAI-assisted learning. CU indeed represents a core dimension of AI literacy, supporting students in developing the judgement, evaluative practices, and self-regulatory skills necessary for navigating AI-enhanced academic

environments. By encouraging students to interrogate AI outputs, validate ideas independently, and consider the appropriateness of GenAI across different learning contexts, CU safeguards against over-reliant usage.

## 5.2. Implications

This study employs PLS-SEM as a predictive educational data analytics approach to investigate students' critical use and reliance on GenAI across four functional domains: understanding, assessment, programming, and engineering project work. When integrated with an extended Technology Acceptance Model (TAM), PLS-SEM enables the modelling of complex latent relationships among multiple predictors, offering both explanatory and predictive insights into GenAI-related behaviours. Its predictive capability is particularly valuable for examining how students' acceptance and evaluative engagement translate into different reliance patterns across academic tasks. The study highlights the importance of moving beyond initial technological acceptance to examine post-adoption behaviours, such as critical use and reliance, particularly in technically demanding disciplines.

The findings show that PEU, PU, ATT and CU collectively shape students' intention to use GenAI, which subsequently predicts their reliance across the four domains. These emphasise the importance of modelling post-adoption behaviours beyond initial technological acceptance. For higher education institutions, such predictive insights can inform evidence-based deployment, allowing educators to anticipate where reliance is likely to emerge and to distinguish between critical and potentially uncritical use. Academic staff can draw on these insights to demonstrate how GenAI may enhance efficiency and learning support without undermining critical engagement through instruction, training sessions, or workshops. Educators, therefore, play a key role in guiding the use of GenAI as a learning support tool while ensuring it does not undermine students' analytical reasoning, problem-solving, or critical thinking (Symeou et al., 2025; Wang et al., 2025).

AI literacy training in Engineering and CS/IT programs could adopt a three-tiered approach. At the first tier, all students should receive evidence-based AI literacy training covering cognitive, affective and socio-cultural dimensions (Kong & Zhang, 2021). This should include teaching foundational knowledge of AI concepts and applications, empowering students in using AI, and enhancing awareness of ethical issues (Kong et al., 2021). Institutions should support this training through access to approved AI platforms, monitored learning environments, and clear guidelines on acceptable use to promote ethical and equitable engagement (Moorhouse et al., 2023; Tillmanns et al., 2025). Professional development for teaching staff may also facilitate pedagogically aligned GenAI use. At the second tier, students needing additional assistance could be referred to mentoring or small-group targeted support. Self-assessments, self-reflections, and discipline-specific AI literacy tests could be developed to identify students at risk of overreliance or uncritical use and guide early intervention (Chen & Zhang, 2025). The third tier should involve more intensive support for students experiencing persistent difficulties, including overdependence on GenAI, low confidence, or limited critical evaluation skills. This may involve one-to-one consultations, personalised learning plans, and regular mentoring.

The findings suggest that students use GenAI differently across academic tasks; therefore, pedagogical approaches should be task-specific rather than uniform. Initial scaffolding may model how to use GenAI to generate explanations, solutions, or code, followed by guidance to evaluate errors and risks. This scaffolding can then be gradually reduced as appropriate. Ongoing monitoring is important to identify changing patterns of AI use and determine whether interventions are strengthening students' critical engagement and independent problem-solving.

## 5.3. Limitations and future research

While this study offers important insights, several limitations should be acknowledged. The research was conducted with a small sample drawn from an Australian university ( $n = 126$ ), which satisfied the minimum sample size for this PLS-SEM model ( $n = 124$ ), as calculated by the minimum R-squared method (power = 0.8; significance level = 0.05) (Kock & Hadaya, 2018). Although the findings provide a useful snapshot of engineering and IT students' engagement with GenAI, the institutional and disciplinary specificity of the sample may limit the generalisability of the results. The findings may therefore not fully represent students in other disciplines, institutions, or national contexts. Broader and more diverse sampling is therefore needed in future studies to help validate and extend the present findings. With increased sample sizes, analytical approaches such as latent profile analysis (Masyn, 2013) could be applied to classify latent classes of students with similar GenAI behaviour patterns. In addition, multigroup analysis (such as by educational level, study field, age and gender) could be conducted to establish if the effects vary across groups. Larger-scale studies should also be extended to capture other emerging aspects of GenAI, such as satisfaction and academic performance and associated factors.

The study relied on self-report survey data, which carries an inherent risk of response bias. Despite measures taken to ensure clarity and anonymity, participants may still have underreported their actual use and reliance on GenAI due to social desirability tendencies, especially when describing behaviours that could be perceived as academically inappropriate. Such bias may influence the accuracy and interpretation of the constructs examined. Future studies incorporating in-depth interviews, focus groups, or classroom observations could triangulate survey findings. Future research could also collect longitudinal data to examine how students' critical use and reliance on GenAI tools evolve as GenAI tools develop.

## 6. Conclusion

This study has extended the TAM framework by integrating critical use to discover factors influencing Engineering and CS/IT students' intention to use and reliance on GenAI tools. The results of the PLS-SEM model demonstrated that while perceived ease of use and perceived usefulness had a positive indirect influence on students' intention to use GenAI, attitudes and critical use directly and positively predicted the intention. Moreover, the intention, in turn, positively predicted reliance on GenAI across four functional domains: understanding, assessment, programming, and engineering projects. Importantly, students' reliance varies across functional domains. Overall reliance is moderate, with the highest reliance observed for understanding-related activities and programming, and lower reliance for engineering projects and assessment tasks, indicating selective and cautious use. By revealing the nature and patterns of students' intention and reliance behaviours, this study provides practical guidance for higher education institutions aiming to cultivate ethical, critical engagement and appropriate reliance on GenAI in support of learning.

## CRedit authorship contribution statement

**Trang H. Nguyen:** Writing – original draft, Methodology, Investigation, Conceptualization. **Long T. Truong:** Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Nhu H.T. Nguyen:** Writing – review & editing, Methodology, Conceptualization.

## Statements on open data and ethics

The study was approved by an ethical committee with ID: HEC25057. Informed consent was obtained from all participants, and their privacy rights were strictly observed. The data can be obtained by



sending request e-mails to the corresponding author.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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